

SMART LENDER – LOAN ELIGIBILITY PREDICTION SYSTEM

METADATA

* **Date:** 7 July 2026

* **Team ID:** xxxx

* **Project Name:** Smart Lender

* **Maximum Marks:** 3 Marks

* **Team Leader:** Nikhil Kumar Reddy

* **Team Members:** Satish Babu, Naga Venkata Sai Bhavani Kolla, Pavani Kurapati, Gorripati Thanusha Divya Surya

* **Mentor:** No Mentor Assigned

1. Submission Checklist (Step 1)

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files (if applicable)	N/A
5	Environment configuration file (.env.example or similar)	N/A
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	N/A
8	Dockerfile / Containerization config (if applicable)	N/A
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

2. File / Folder Structure (Step 2)

Smart_Lender_Loan_Eligibility/

```
|
|└─ app.py          # Main Flask application entrypoint
|└─ train_model.py   # Preprocessing & ML model training script
|└─ config.py        # Environment setup variables
|└─ requirements.txt  # Project library dependencies
|└─ README.md        # Setup manual and descriptions
|└─ .gitignore       # Version control exclusions
|└─ .python-version   # Python version pin for Render deployment
|
|└─ dataset/         # Dataset source files
|  └─ loan_data.csv   # Raw applicant entries (614 rows)
|
|└─ model/           # Serialized machine learning models
|  └─ loan_model.pkl   # Saved StandardScaler & XGBoost binary
|
|└─ templates/       # HTML templates
|  └─ index.html       # Unified Glassmorphic predictive web form
|
└─ static/           # Asset folders
   └─ css/
      └─ style.css     # CSS styles with visual theme configurations
   └─ js/
      └─ main.js       # Range-slider sync and in-place calculations script
   └─ screenshots/
      └─ class_distribution.png
      └─ confusion_matrix.png
      └─ correlation_heatmap.png
```

3. Deployment / Access Details (Step 3)

Field	Details
Hosted / Deployed URL	https://smart-lender-loan-eligibility-prediction-a7rg.onrender.com
Login Credentials (Demo)	Not Required (Application does not require authentication)
Platform / Hosting Provider	Render
Repository Link	https://github.com/Nikhil-tech664/Smart-Lender-Loan-Eligibility-Prediction
Demo Video Link	https://drive.google.com/file/d/187lrm_ql02IL6k_9gZNwz13sv0fGCNhQ/view?usp=drive_link

4. Run Instructions (Step 4)

1. Clone the GitHub repository:

```
```bash
git clone https://github.com/Nikhil-tech664/Smart-Lender-Loan-Eligibility-Prediction.git
```
```

2. Navigate to the project folder:

```
```bash
cd Smart-Lender-Loan-Eligibility-Prediction
```
```

3. Create a virtual environment:

```
```bash
python -m venv venv
```
```

4. Activate the virtual environment:

```
***Windows:** `venv\Scripts\activate`
***Linux / macOS:** `source venv/bin/activate`
```

5. Install the required dependencies:

```
```bash
```

```
pip install -r requirements.txt
```

```
'''
```

6. Ensure the dataset (`loan\_data.csv`) and trained model binary (`loan\_model.pkl`) are available in their respective folders.

7. Run the Flask application:

```
```bash
```

```
python app.py
```

```
'''
```

8. Open your browser and visit:

```
`http://127.0.0.1:5000`
```

9. Enter applicant details and click ****Predict Eligibility**** to view the prediction result.

5. Known Issues / Limitations (Step 5)

S.No	Known Issue / Limitation	Workaround / Status
1	The application currently uses a CSV dataset and serialized model files instead of a database.	Planned database integration (PostgreSQL) in a future release.
2	User authentication and account management are not implemented.	Planned as a future enhancement.
3	The application is designed for small to medium workloads and has not been optimized for high concurrent user traffic.	Can be improved by using Docker containers, load balancing, caching, and cloud scaling.